

MODULE 4

Tidal Power: Tides and waves as energy suppliers and their mechanics; fundamental characteristics of tidal power, harnessing tidal energy, advantages and limitations.

Ocean Thermal Energy Conversion: Principle of working, OTEC power stations in the world, problems associated with OTEC.

TIDAL POWER:

Tides and waves as energy suppliers and their mechanics:

- Tide is a periodic rise and fall of the water level of sea which are carried by the action of the sun and moon on the water of the earth.
- Tide energy contributes for a significant portion of all renewable energies.
- The large scale up and down movement of sea water is an unlimited source of energy.
- The main feature of the tidal cycle is the difference in water surface elevations at the high tide and at the low tide.
- If this differential head could be utilized in operating a hydraulic turbine, the tidal energy could be converted into electrical energy by means of an attached generator.
- At the time of high tide, water is at a high level and can be let into a basin to be stored at a high level there.
- The same water can be let back into the sea during the low tide through the turbines, thus producing power.
- Since the basin water level is high and sea water is low, there is a differential head comparable to the tidal range, that can be utilized for the running of the turbines.
- Since tides come at varying times from day to day and have varying ranges (heads), it requires large capital expenditures for large outputs.

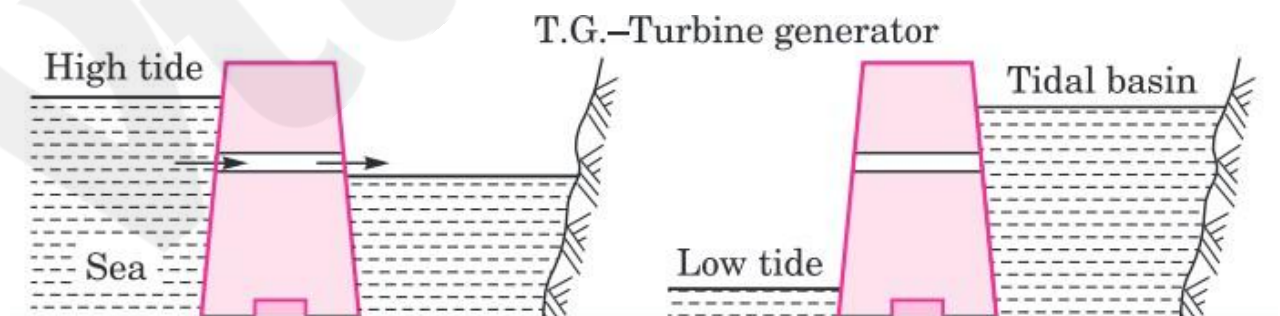


Fig 1: Principle of tidal power

- Tides are produced by the gravitational attraction of the moon and the sun on the water of solid earth and the oceans.
- About 70% of the tide producing force is due to the moon and 30% due to the sun.
- The moon is thus the major factor in the tide formation.

- Surface water is pulled away from the earth on the side facing the moon, and at the same time the solid earth is pulled away from the water on the opposite side.
- Thus, high tides occur in these two areas with low tides at intermediate points.
- As the earth rotates, the position of a given area relative to the moon changes, and so also do the tides – this results in a periodic succession of high and low tides.
- Two tidal cycles (two high tides and two low tides) occur during a lunar day of 24 hours and 50 minutes. These are called semidiurnal tides.
- The lunar day is the apparent time of revolution of the moon about the earth.
- The time between high tides and low tide at any given location is a little over 6 hours.
- A high tide will be experienced at a point which is directly under the moon.
- A diametrically opposite point on the earth's surface also experiences a high tide due to dynamic balancing.
- Thus, a full moon as well as a no moon produce a high tide.
- The rise and fall of the water level follows a sinusoidal curve [Fig 2].

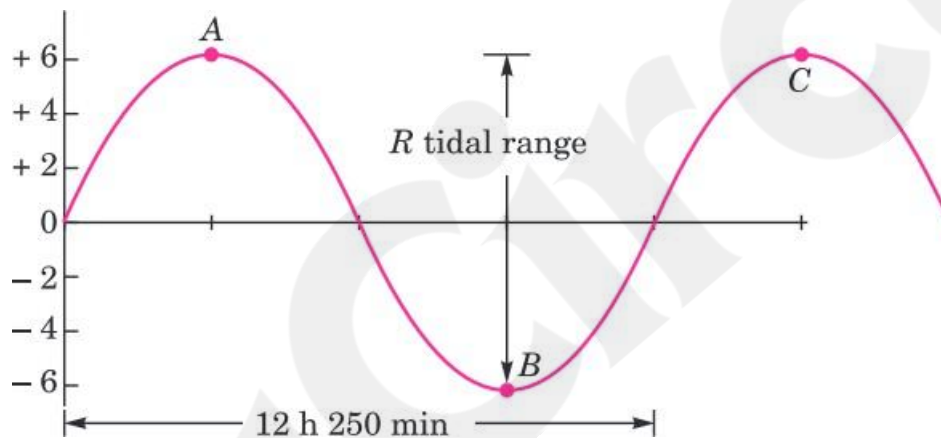


Fig 2: Tides of Sea

- The difference between high and low water levels is called the range of the tide.
- Tidal range is defined as $R = \text{water elevation at high tide} - \text{water elevation at low tide}$.
- Because of the changing positions of the moon and sun relative to the earth, the range varies continuously.
- At times near full or new moon, when sun, moon and earth are approximately in a line, the gravitational forces of sun and moon enhance each other, and the tidal range is then exceptionally large (high tides are higher and low tides are lower than the average) - These high tides are called **spring tides**.
- Near the first and third quarters of the moon, when the sun and moon are at right angles with respect to the earth, **neap tides** occur. The tidal range is exceptionally small (high tides are lower and the low tides higher than the average).
- Thus, the tidal range is not constant and varies during the 29.5 day lunar month [Fig. 3]
- The spring neap tidal cycle lasts one-half of a lunar month.
- A typical mean range is roughly one third of the spring range.

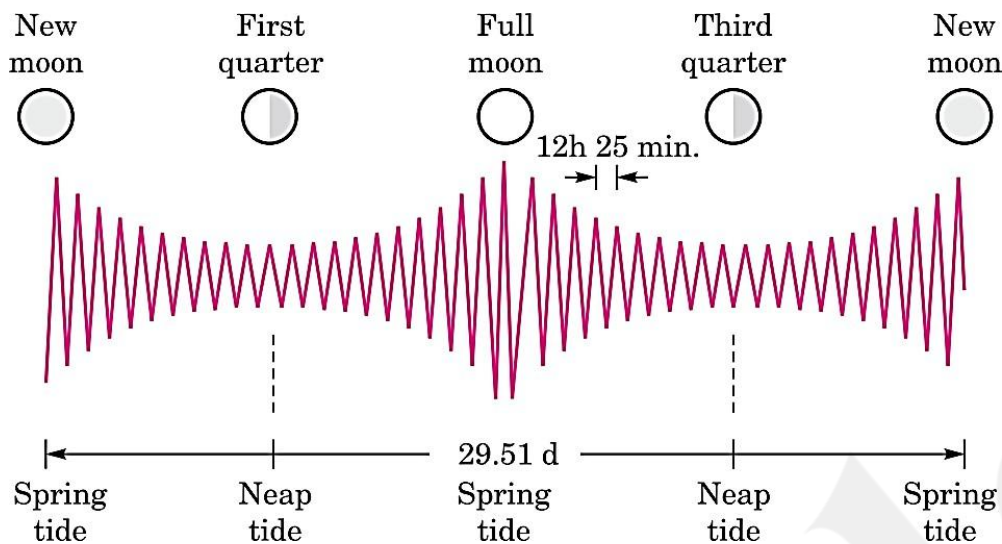


Fig 3: Variations of the tidal range

Components of Tidal power plant:

The three main components of a tidal power plant are:

- (i) The powerhouse
- (ii) The dam or barrage (low wall) to form pool or basin
- (iii) Sluice ways from the basins to the sea and vice versa

Dam (Barrage):

- The function of dam is to form a barrier between basin and sea.
- The powerhouse as well as the sluice ways should be in alignment with the dam.
- Tidal power barrages must resist waves whose shock can be severe and where pressure changes sides continuously.
- The barrage needs to provide channels for the turbines in prestressed or reinforced concrete.
- Prefabricated concrete blocks can be used as the core for large barrages and voids filled with rocks or concrete remaining holes with sand, and the entire construction then asphalted.
- Construction of a barrage will influence the tidal amplitude.
- Tidal barrages require sites where there is a sufficiently high tidal range to give a good head of water.
- The location of the barrage is important, because the energy available is related to the size of the trapped basin and to the square of the tidal range.

Gates and Locks:

- Tidal power basins must be filled and emptied.
- Gates are opened regularly and frequently; It must be opened and closed rapidly so that this operation should use a minimum of power.
- The design also provides for pumping between the basin and the sea in either direction.
- If reversible pump turbines are provided, the pumping operation can be taken over at any time by the same machine.

- The modern tubular turbines are so versatile that they can be used either as turbines or as pumps in either direction of flow.
- In addition, the tubular passages can also be used as sluice ways by locking the machine to a standstill.
- In existing plants, vertical lift gates have been used; It can be substituted with a series of flaps that operate by water pressure.
- The flap gates allow only in the direction of the sea to basin; Hence, the basin level rises well above to sea level.

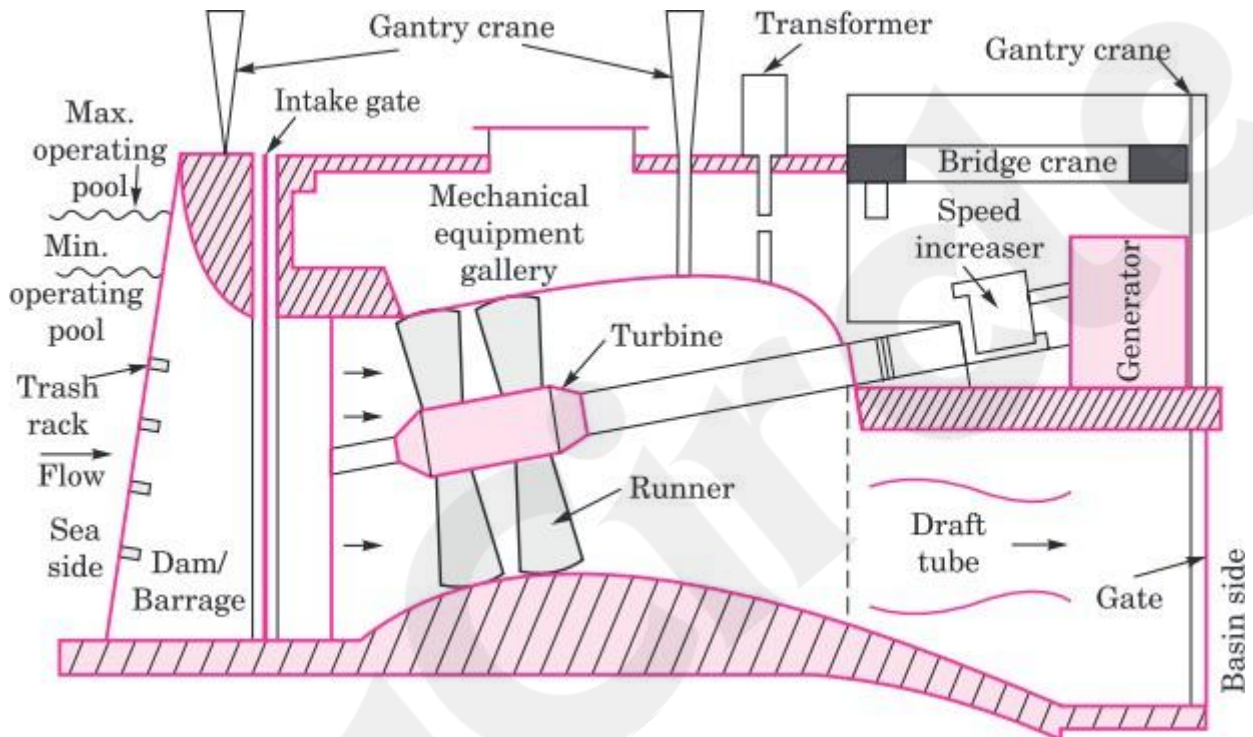


Fig 4: Schematic layout of Tidal power plant

Powerhouse:

- The turbines, electric generators and other auxiliary apparatus are the main equipment of a powerhouse.
- For small water heads, large size turbines are needed which makes the powerhouse a large structure.
- Both the French and Soviet operating plants use the bulb type of turbine.
- Of the propeller type, with reversible blades, bulbs have horizontal shafts coupled to a single generator.
- The cost per installed kilowatt drops with turbine size.
- **A bulb type turbine** – It is an axial flow turbine; the bulb set which resembles a small submarine, is made up of an ogive shaped steel shell containing an alternator and a Kaplan turbine. It is placed in a horizontal hydraulic duct and surrounded by water; a shaft provides communication with the engine room of the power plant. The set functions as a turbine and as a pump, and regulates the flow in both directions of flow, tide to reservoir (basin) and reservoir to tide.

- **Rim Type Turbines** – In rim turbine, the rotor surrounds the turbine runner i.e., the rim is carried by the runner blades; the generator is attached peripherally to the turbine blades, an arrangement that couples two turbines of conventional type to one generator, and a hydraulic system in which up to six turbines are coupled to hydrostatic pumps and to drive a Pelton-wheel, which, in turn, drives a high-speed generator.

Fundamental characteristics of tidal power:

- (1) The tides are a periodical phenomenon but no two tides in any cycle are alike. Since the relative positions of sun and moon and their distances from earth are continuously changing, the tides are also influenced accordingly. Of the two high tides in a single day, one tide is higher than the other
- (2) The mean tidal range varies from place to place. The shape of the tidal cycle depends upon the interaction of the sea with the coastline. Where the coastline offers a resonating influence, the tidal range gets accentuated, at the other places, the land may produce a dampening effect on the tidal phenomenon.
- (3) In spite of their complexity, the tides are amenable to mathematical analysis. As a result, the exact time and the water level for a high tide as low tide can be forecast with great accuracy.

Harnessing tidal energy:

- The generation of electricity from waterpower requires a difference in levels (or heads) between which water flows.
- Several concepts have been proposed for generating electricity by utilizing the head that can be produced by the rise and fall of the tides to operate a hydraulic turbine.
- The power generation from tides involves flow between an artificially developed basin and the sea.
- To have a continuous generation, this basic scheme can be elaborated by having two or more basins. Accordingly, there are two types of arrangements:

(1) Single basin arrangement

(2) Double basin arrangement

(1) Single Basin Arrangement:

- In a single basin arrangement, there is only one basin interacting with the sea.
- The sea and basin are separated by a dam (or barrage) and the flow between them is through sluice ways located conveniently along the dam.
- Potential head is provided by rise and fall of tidal water levels which is accomplished by blocking the mouth of a long narrow estuary with a dam across it, thereby creating a reservoir.
- The dam or barrage embodies several sluice gates and low head turbine sets.
- The generation of power can be achieved in a single basin arrangement as a
 - (a) Single ebb cycle system, or
 - (b) Single tide cycle system, or
 - (c) Double cycle system.

(a) Single ebb cycle system:

- When the flood tide (high tide) comes in, the sluice gates are opened to permit seawater to enter the basin or reservoir, while the turbine sets are shut.
- The reservoir thus starts filling while its level rises, till the maximum tide level is reached.
- At the beginning of the ebb tide the sluice gates are closed.
- Then the generation of power takes place when the sea is ebbing (flowing back of tide) and the water from the basin flows over the turbines into the lower-level sea water.
- After two or three hours when there is sufficient difference between the full reservoir level and the falling tide level, to run the turbines, they are started.
- The turbine keeps working until the rising level of the next flood tide and the falling reservoir level together reduce the effective head on the turbines to the extent where it can no longer work safely and efficiently.
- The turbines are then closed and the sluice gates opened again to repeat the cycle of operations.
- Since in an estuary, the ebb tide has a longer duration than the flood tide, the ebb operation provides an increased period of actual work.

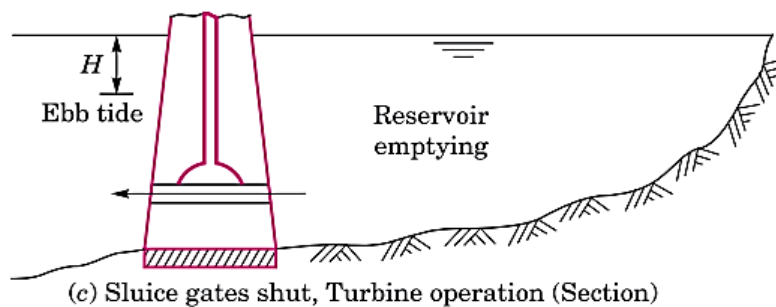
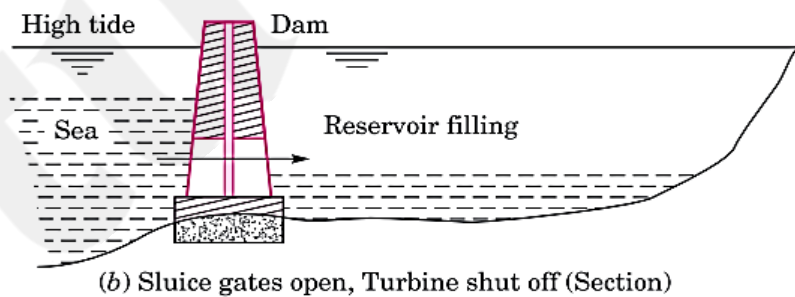
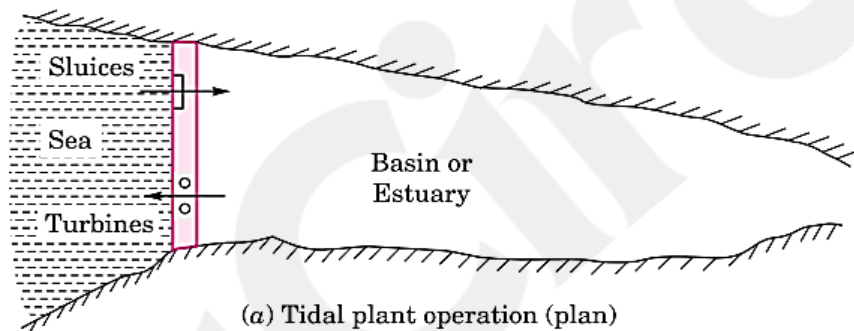


Fig 5: Tidal power plant – Single basin operation

(b) Single tide cycle system:

- In a single tide cycle system, the generation is affected when the sea is at flood tide.
- The water of the sea is admitted into the basin over the turbines.
- When the flood tide period is over and the sea level starts falling again, the generation is stopped.
- The basin is drained into the sea through the sluice ways.
- Flood tide operation scheme needs larger size plant, operating for shorter period and hence less efficient as compared to ebb tide operation.
- The ebb operation plant will be of smaller size but will operate over a large period.
- The aim should be to obtain as long a period of operation as possible at the beginning and finishing the work at the minimum operating head.
- Energy produced by an ebb cycle system is 1.5 times that by a tide cycle system.
- The main disadvantage in both the ebb cycle as well as the tide cycle systems is the intermittent nature of their operation.
- Since the intermissions occur at regular intervals, there is possibility of connecting another supplementary system to balance the discontinuity and regulate the output.

(c) Double cycle system:

- The system can be geared to generate power both during the ebb and flood tides with the help of single basin only.
- This system is known as the double cycle system.
- Power generation is accomplished both during emptying and filling cycles.
- Both filling and emptying processes take place during short periods of time.
- Filling happens when the ocean is at high tide while the water in the basin is at low tide level.
- Emptying happens when the ocean is at low tide and the basin at high-tide level.
- The flow of water in both directions is used to drive several reversible water turbines, each driving an electrical generator.
- Electric power would thus be generated during two short period during each tidal period of 12 h, 25 min. or once every 6h, 12.5 min.

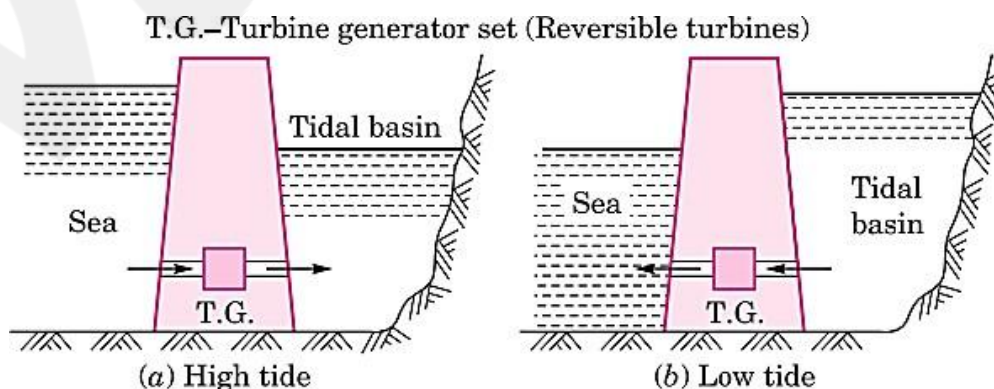


Fig 6: Double cycle system

- Though the double cycle system has only short duration interruptions in the turbine operation, yet a continuous generation of power is still not possible.
- Furthermore, the periods of power generation coincide only occasionally with periods of peak demand.
- These problems are solved to some extent in the two-basin scheme.
- However, a fundamental drawback to all methods for generating tidal power is the variability in output caused by the variations in the tidal range.

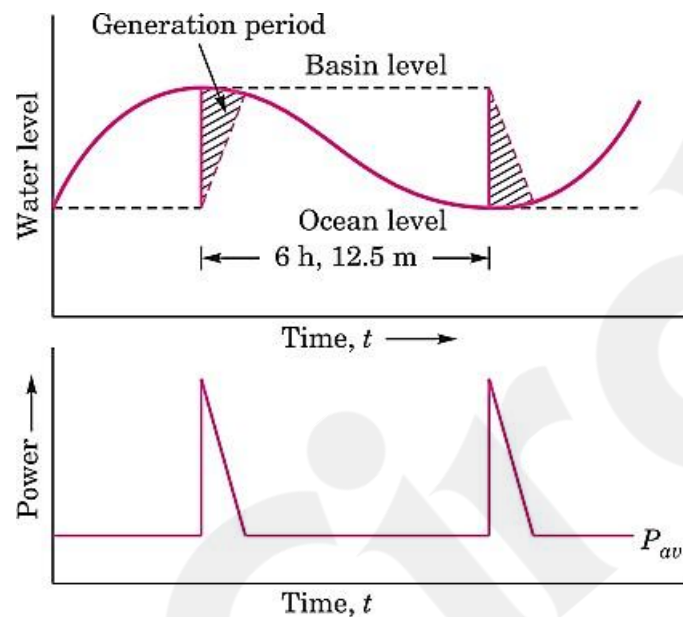


Fig 7: Ocean and pool levels and power generated in a single basin tidal system

(2) Double Basin Arrangement:

- In this system the turbines are located between the two adjacent basins, while the sluice gates are as usual embodied in the dam across the mouths of the two estuaries.
- At the beginning of the flood tide, the turbines are shut down, the gates of upper basin A are opened and those of the lower basin B are closed.
- The basin A is thus filled up while the basin B remains empty.
- As soon as the rising water level in A provides sufficient difference of head between the two basins, the turbines are started.
- The water flows from A to B through the turbines, generating power.
- The power generation thus continues simultaneously with the filling up the basin A.
- At the end of the flood tide when A is full and the water level in it is the maximum, its sluice gates are closed.
- When the ebb tide level gets lower than the water level in B, its sluice gates are opened whereby the water level in B, which was rising and reducing the operating head, starts falling with the ebb.
- This continues until the head and water level in A is sufficient to run the turbines.
- With the next flood tide the cycle repeats itself.

- With twin basin system, a longer and more continuous period of generation per day is possible.
- The small gaps in the operation of such stations can be filled up by thermal power.

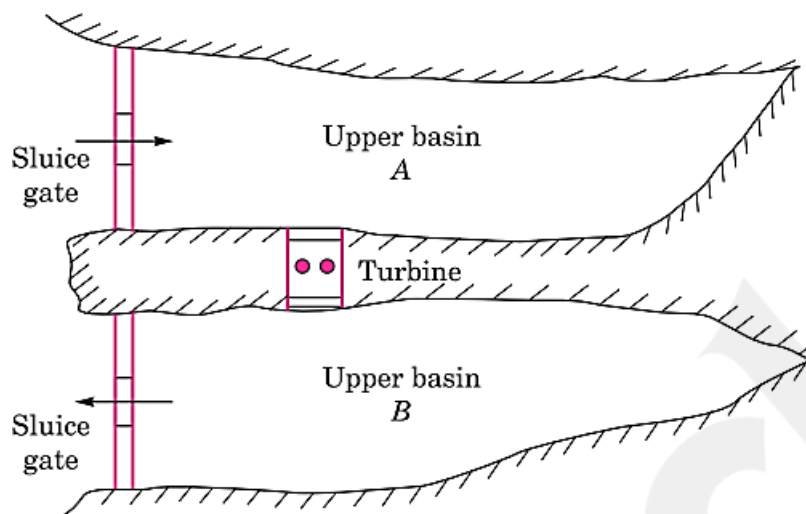


Fig 8: Tidal power plant – Double basin operation

- The operation of the two-basin scheme can be controlled so that there is a continuous water flow from upper to lower basin.
- Since the water head between the basins varies during each tidal cycle, as well as from day to day, so also does the power generated.
- As in the case with single basin scheme, the peak power generation does not often correspond in time with the peak demand.
- One way of improving the situation is to use off-peak power from the tidal power generators or from an alternative system to pump water from the low basin to the high basin.
- Thus, an increased head will be available for power generation during peak demand.
- This is very similar to pumped storage system in hydro-electric power stations.

Advantages and Limitations:

Advantages:

- (1) Tidal power is **inexhaustible**, and it is completely independent of the precipitation (rain) and its uncertainty; Even a continuous dry spell of any number of years can have no effect on the tidal power generation.
- (2) Tidal power generation is **free from pollution**, as it does not use any fuel and does not produce any unhealthy waste like gases, ash, and atomic refuse.
- (3) Tidal power plants **do not demand large area of valuable land** because they are on the bays (seashore).
- (4) **Peak power demand can be effectively met** when it works in combination with thermal or hydroelectric system.

Limitations:

- (1) The fundamental drawback to all methods of generating tidal power is the variability in output caused by the variations in the tidal range.
- (2) The tidal ranges are highly variable and thus the turbines must work on a wide range of head variations. This affects the efficiency of the plant.
- (3) Since the tidal power generation depends upon the level difference in the sea and an inland basin, it must be an intermittent operation, feasible only at a certain stage of the tidal cycle. This intermittent pattern could be improved to some extent by using multiple basins and a double cycle system.
- (4) The tidal range is limited to a few meters. Bulb turbine technology was not well developed and the use of conventional Kaplan runners was the only alternative. This was found to be unsuitable. Now with the development of reversible flow bulb turbines, this difficulty is overcome.
- (5) The duration of power cycle may be reasonably constant, but its time of occurrence keeps changing, introducing difficulties in the planning of the load sharing every day in a grid. This is removed now with the help of computerized programming.
- (6) Sea water is corrosive, and it was feared that the machinery may get corroded. Stainless steel with a high chromium content and a small amount of molybdenum and the aluminum bronzes proved to be good corrosion resistant. The vinyl paint exhibited good results.

OCEAN THERMAL ENERGY CONVERSION:

Principle of working:

- The operation of the OTEC plant is based on thermodynamic principle.
- If a heat source is available at a higher temperature and a heat sink at a lower temperature, the temperature difference can be utilized in a turbine which converts part of the heat into mechanical energy and hence into electrical energy.
- The residual heat is discharged to the sink at the lower temperature.
- In the OTEC system, the warm ocean surface water is the heat source, and the deep colder water provides the sink.
- This is called ocean thermal energy conversion (OTEC).
- OTEC system has very low efficiency and has very high capital cost, because the temperature difference is small even in tropics.
- The heat contained in the ocean is solar in origin.
- The surface of the water acts as the collector for solar heat while the upper layer of the sea constitutes infinite heat storage reservoir.
- Solar energy absorption by the water takes place according to Lambert's law of absorption, which states that each layer of equal thickness absorbs the same fraction of light that passes through it.
- Mathematically,

$$-\frac{dI_{(x)}}{dx} = kI$$
$$I_{(x)} = I_0 e^{-kx}$$

- where, I_0 and $I(x)$ are the intensities of radiation at the surface ($x = 0$) and at a distance x below the surface.
- K is an extinction coefficient (or absorption coefficient); K has values of 0.05 m^{-1} for very clear fresh water, 0.27 for turbid fresh water and 0.50 m^{-1} for very salty water.
- Thus, the intensity decreases exponentially with depth and depending upon K , almost all the absorption occurs very close to the surface of deep waters.
- In the tropics, the ocean surface temperature often exceeds 25°C , while 1 km below, the temperature is usually not higher than 10°C .
- In tropical waters there are two infinite heat reservoirs, a heat source at the surface at about 27°C and a heat sink some 1 km directly below, at about 4°C .
- The concept of ocean thermal energy conversion (OTEC) is based on the utilization of this temperature difference in a heat engine to generate power.
- The surface temperatures and temperature differences vary both with latitude and season, both being maximum in tropical, subtropical, and equatorial waters.

OTEC power stations in the world:

Operational OTEC plants:

1. Okinawa, Japan

Plant Name: Okinawa OTEC Demonstration Facility

Capacity: 100 kW

Status: Operational (since 2013)

Details: Located on Kume Island, this is the world's only continuously operating OTEC plant. It's used for research, power generation, and supporting local aquaculture.

2. NELHA, Hawaii, USA

Plant Name: Mini-OTEC (historical) / Later demonstration systems

Capacity: 50–250 kW (demonstration projects)

Status: Decommissioned / Research and development continue

Notes: The Natural Energy Laboratory of Hawaii Authority (NELHA) has hosted several OTEC experiments, including the first successful closed-cycle OTEC plant in 1979. The 250 KW demonstration plant supplies electricity to the local grid. Makai Ocean Engineering also operates a larger OTEC plant in Hawaii, which is the world's biggest operational plant.

3. Kavaratti, Lakshadweep, India

The National Institute of Ocean Technology (NIOT) is establishing a 65 kW OTEC plant to power a low-temperature thermal desalination plant, converting seawater into potable water.

Proposed / Under Development OTEC Projects:

3. Martinique, Caribbean (France)

Project Name: NEMO (New Energy for Martinique and Overseas)

Capacity: Initially planned for 10.7 MW

Status: On hold / Cancelled (as of latest reports)

Notes: Led by DCNS (now Naval Group) and Akuo Energy; faced economic and logistical challenges.

4. Maldives

Capacity: ~1 MW (proposed)

Status: Feasibility studies / development stages

Notes: Government has explored OTEC as part of its sustainable energy plan for island resilience.

5. India (Lakshadweep and Andaman Islands)

Status: Conceptual stage

Notes: The Indian government and institutions like NIOT (National Institute of Ocean Technology) have shown interest in OTEC as part of remote island electrification.

6. Indonesia

Status: Feasibility studies

Notes: Indonesia's vast oceanic territory makes it a promising candidate for future OTEC developments.

Methods of OTEC power generation:

There are two different methods for harnessing ocean thermal difference:

- (i) Open cycle system (Claude cycle)
- (ii) Closed cycle system (Anderson cycle)

(i) Open cycle OTEC System (Claude cycle):

- In the open cycle turbine system, water is the working fluid.
- The warm surface water is caused to boil by lowering the pressure, without supplying any additional heat.
- The low-pressure steam produced then drives a turbine, and the exhaust steam is condensed by the deep colder water and is discarded.
- This cycle is named as 'open' because the condensate need not be returned to the evaporator.
- Instead, the condensate, can be utilized as desalinated water.
- In the cycle shown in Fig. 9, warm surface water at say 27°C is admitted into an evaporator in which the pressure is maintained at a value slightly below the saturation pressure corresponding to that water temperature.
- At the new pressure, water which is entering the evaporator gets 'superheated'.
- The low pressure in the evaporator is maintained by a vacuum pump that also removes the dissolved non condensable gases from the evaporator.

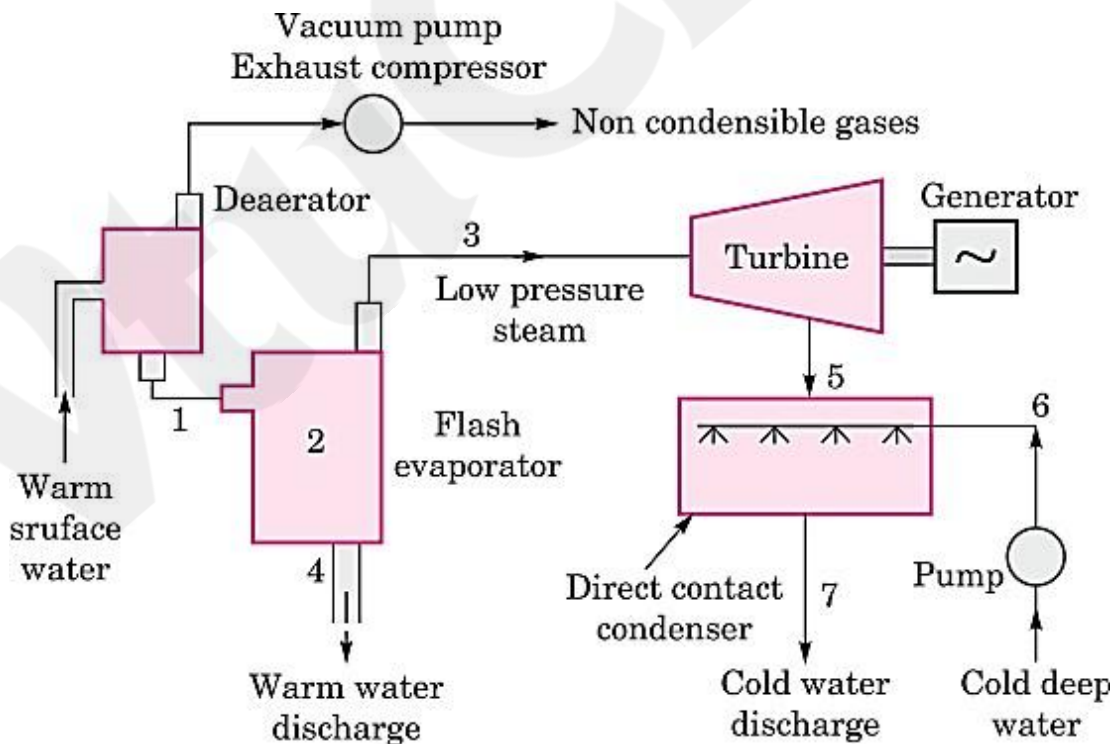


Fig 9: Schematic of the Open cycle OTEC system

- As shown in Fig. 10, the warm water which is at 27°C, has a saturation pressure of 0.03619 kg/cm² (0.0356 bar) (point 1).
- The evaporator pressure is 0.03213 (0.0317 bar), which corresponds to 25°C saturation temperature.
- This temporarily superheated water undergoes volume boiling causing that water to partially flash to steam to an equilibrium two phase condition at the new pressure and temperature of 0.03213 kg/cm² and 25°C (point 2).
- Process 1-2 is a throttling and hence constant enthalpy process.
- At point 2, the evaporator contains a mixture of water and steam of very low quality.
- The steam is separated from the water as saturated vapour at 3.
- The remaining water is saturated at 4 and is discharged as brine back to the ocean.
- The steam at 3, has a very low pressure and high specific volume (0.03213 kg/cm², 43.40 m³/kg), as compared to conventional fossil power plant.
- In the condenser, the exhaust steam is mixed with cold water from the deep cold-water pipe at 6, which results in a near saturated water at 7.
- This water is allowed to be discharged to the ocean.
- Cooling water (11°C) from the deep ocean on reaching the condenser rises to about 15°C due to heat transfer between the warmer outside water and cooling water inside the pipe, as it ascends towards the top.
- Very large ocean water mass and volume flow rates are used in open OTEC systems.
- To harness the energy in low pressure steam, extremely large turbines must be used.
- De-gasifiers (deaerators) must be used to remove the gases dissolved in the sea water otherwise efficiency will be reduced.
- Since there are no heat transfer problems in the evaporator, the problem of bio-fouling control is minimized.
- The cost of an open cycle system is significantly greater than for closed cycle system.
- The turbine cost constituted almost half the cost of the power system.

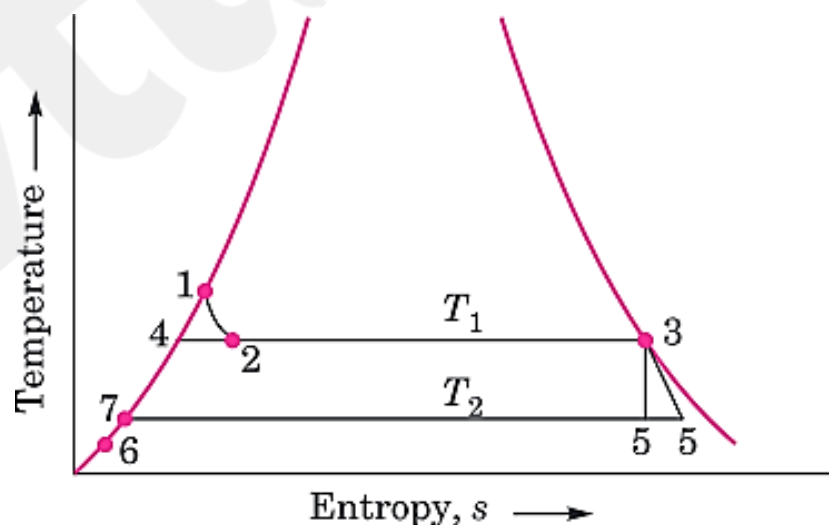


Fig 10: T-S diagram of Open cycle system

(ii) Closed cycle OTEC System:

- In the closed cycle system, a liquid working fluid, such as Ammonia, Propane or Freon is vaporized in an evaporator (or boiler).
- The heat required for vaporization is transferred from the warm ocean surface to the liquid by means of a heat exchanger (evaporator and condenser).
- The high-pressure vapour leaving the evaporator drives an expansion turbine, like a steam turbine that it is designed to operate at a lower inlet pressure.
- The turbine is connected to an electric generator in the usual manner.
- The low-pressure exhaust from the turbine is cooled and converted back into liquid in the condenser.
- The cooling is achieved by passing cold, deep ocean water, from a depth of 700 to 900 m or more, through a heat exchanger.
- The liquid working fluid is then pumped back as high-pressure liquid to the evaporator, thus closing the cycle.
- The operating (saturation) pressures of ammonia, propane or Freon at the boiler and condenser temperatures are much higher than those of water, being roughly 10 kg/cm² (= 10 bar) at the boiler, and their specific volumes are much lower.
- Such pressures and specific volumes result in turbines that are much smaller and hence less costly than that of the open cycle system.
- The closed cycle also avoids the problems of the evaporator.
- It requires the use of very large heat exchangers (boiler and condenser) because, for an efficiency of about 2%, the amounts of heat added and rejected are about 50 times the output of the plant.

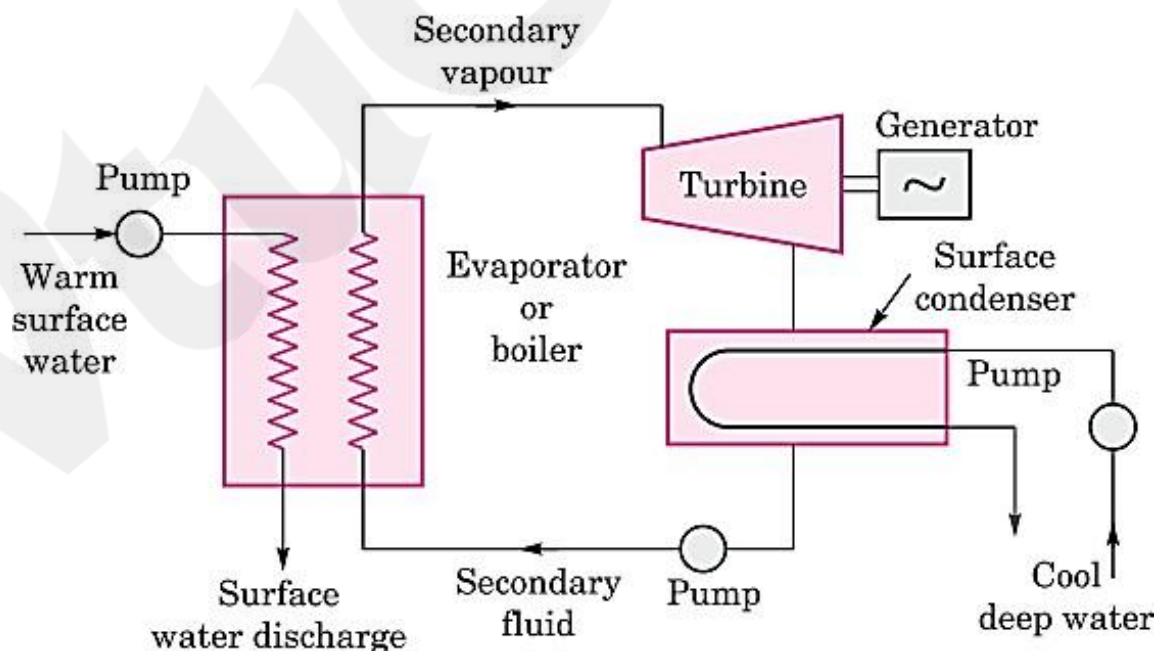


Fig 11: Schematic of Closed cycle OTEC system

(a) Anderson closed cycle OTEC system:

- In the Anderson cycle propane was chosen as the working fluid.
- The temperature difference between warm surface and cool surface was 20°C.
- The cool surface was at about 600 m deep.
- Propane is vaporized in the boiler or evaporator at about 10 kg/cm² (10 bar) or more and exhausted in the condenser at about 5 bars.
- Propane is compatible with most heat exchanger materials, but it is highly flammable and forms an explosive mixture with air.
- Instead of usual heavier and more expensive shell and tube heat exchangers, the Anderson OTEC system employs thin plate type heat exchangers, which minimizes the mass and the amount of material and hence cost.
- The heat exchangers are placed at depths where the static pressure of the water in either heat exchanger roughly equals the pressure of the working fluid, this helps in reducing the thickness of the plates.

(b) Rankine closed cycle OTEC system:

- Rankine cycle uses a working fluid with higher vapour pressure (such as ammonia, hydrocarbon or halocarbon) at the temperature available.
- Heat is transferred sequentially from sea water through the heat exchanger materially (a metallic alloy) and hence to the working fluid (e.g., ammonia).
- For the condenser, an overall U characterized the reverse heat transfer process.
- Ammonia has better operating characteristics than propane and it is much less flammable.
- Ammonia forms a noxious vapour and cannot be used with copper heat exchanger.

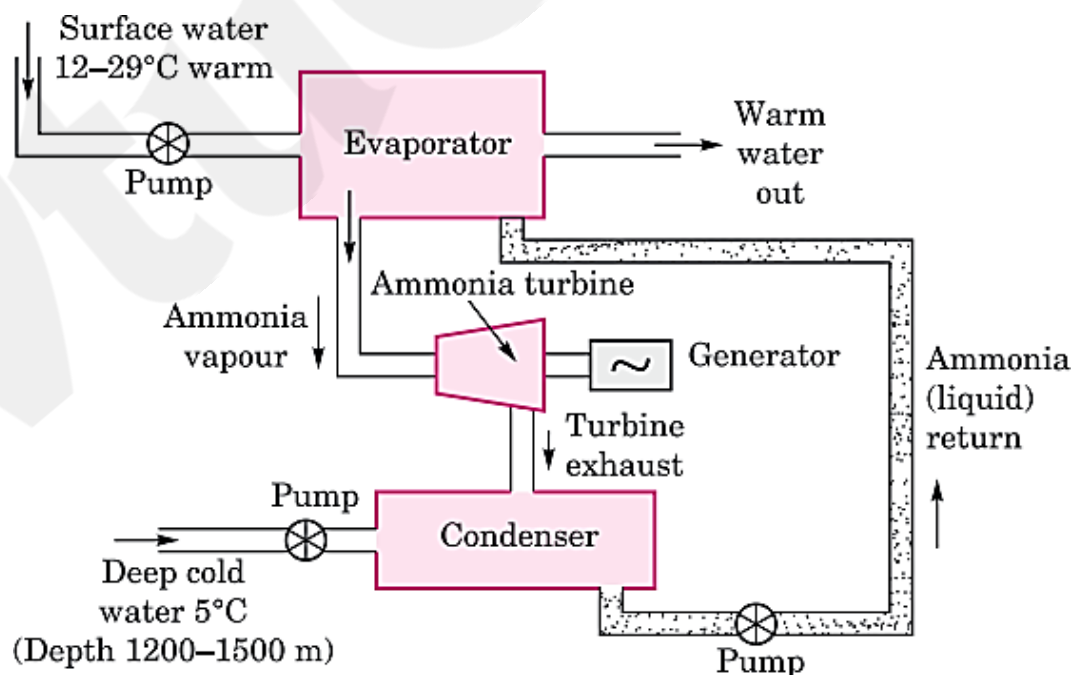


Fig 11: Schematic of Closed OTEC Ammonia cycle system

Problems associated with OTEC:

(1) Heat Exchangers (Evaporators):

- The maximum efficiency for the conversion of heat into mechanical work in a turbine depends on the drop in temperature of the working fluid in its passage through the turbine and the turbine inlet temperature.
- The temperature drop in the turbine is 10°C (283 K) and the inlet temperature is 20°C (293 K), hence the maximum thermal efficiency is $10/273 = 0.034$ or 3.4%.
- Any change from ideal behaviour in the turbine and the extra energy required to pump cold water from great depths will the net efficiency for power generation to 2.5%.
- In the OTEC system, the low conversion efficiency can be compensated by the enormous amounts of heat available in ocean surface waters.
- To utilize this heat to generate electricity in useful amounts, water must be pumped through the heat exchangers in both evaporator and condenser at very high rates.
- The effectiveness and cost of the heat exchangers are critical for the OTEC concept.
- The electric power that can be generated depends on the rate of heat transfer from the warm ocean water to the working fluid in the evaporator.
- Conversion of this heat into electrical energy with maximum efficiency requires that the temperature of the working fluid entering the turbine should be as high as possible and that of the fluid leaving the turbine as low as possible.
- All these requirements can be met only if there is effective heat transfer in the heat exchangers.
- The constructional materials must have good heat conductivity, sufficient mechanical strength and be resistant to corrosion and erosion by rapidly flowing ocean water.
- The best materials are (1) titanium (2) aluminium (or an alloy) (3) an alloy of copper (90%) and nickel, and (4) plastic.
 - Titanium is resistant to corrosion and erosion by ocean water and it has good mechanical strength.
 - Aluminium is cheaper than titanium, but more susceptible to corrosion by ocean water; Aluminium alloys have been found to be suitable for use with salt water. Even with aluminium alloys, heat exchangers had to be replaced more often than those of titanium.
 - A 90/10 copper-nickel alloy has been used extensively in both land based and shipboard power plant condensers with ocean water as coolant. It is less expensive than titanium but costlier than aluminium alloys. Although the copper nickel alloy is resistant to corrosion by sea water, the copper is readily attacked if ammonia is present. Hence, copper heat exchangers could probably not be used with ammonia as the working fluid.
 - Fabricating heat exchangers from plastic material is less expensive; Heat conductivity of plastics is too low for efficient heat exchange but can be increased by inclusion of graphite. Even with the graphite filled plastic, large heat exchange area is necessary for a specified electrical output. Plastics are resistant to corrosion by ocean water and ammonia but affected by propane.

(2) Biofouling:

- In an ocean environment, a layer of slime known as "bio fouling" will eventually accumulate on the water side of the heat exchangers.
- Slime at first stage is comprised of microorganisms, called as "micro fouling".
- Subsequently, if the slime is not removed, additional biofouling in the form of macro-organisms will become attached, augmenting the slime layer.
- Occurrence of microfouling is a prerequisite for the attachment of macro-organisms.
- A film of corrosion and calcareous deposits can also accumulate on the water side.
- The formulation of biofouling, corrosion, and so on is referred to as "fouling" or scaling and will tend to inhibit heat transfer through it.
- The "fouling factor" is a measure of the thermal resistance 'R' of a fouling film.
- This thermal resistance is the reciprocal of the corresponding heat transfer coefficient 'h' of the fouling film.
- For the heat exchangers to be efficient, provisions must be made to inhibit the formation of fouling layers and to remove any significant fouling that forms.
- Removal can be accomplished by periodically cleaning the heat exchanger surfaces through mechanical, chemical or other means.
- By increasing the flow rate of the water, the organisms are less likely to become attached to the heat-exchange surfaces.
- However, the flow rate must not be high enough to cause erosion.
- Biofouling effects and ways of dealing with them are important in the design and location of OTEC plants.
- Such effects are significant for the evaporator heat exchanger where warmer water would be conducive to the growth of marine organisms.

(3) Site Selection:

- In selecting a site for an OTEC plant, the primary consideration is a significant temperature difference (at least about 20°C) between surface and deep ocean waters.
- The greater the difference, the lower the cost of generating electricity will be.
- The best sites are in the tropical belt between about 20°N and 20°S latitude.
- In choosing a site, consideration should be given to the potential for biofouling effects.
- An OTEC plant would be located offshore to provide access to the deep colder water.
- An ideal situation will be the one where the shoreline dropped steeply to a considerable depth so that the installation could then be more conveniently build on land.

(4) Energy Utilization:

- If OTEC plant is located less than about 30 km from shore, electricity generated can be transmitted inexpensively to land by submarine cable.
- If the plant is so far from shore transmission costs increase, hence the electricity generated can be utilized at the plant site to produce energy intensive materials.
- Direct electric current can be used to decompose sea water by the process of electrolysis; the main products would be hydrogen and oxygen.
- Hydrogen can be combined with atmospheric nitrogen to form ammonia for use as fertilizer.